

Second Terminal Examination - 2082

Class : XI	Subject: Biology (Code: 2011)	Full Marks: 75
Time: 3 hrs.	SET 'B'	Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet. [6 × 1 = 6]

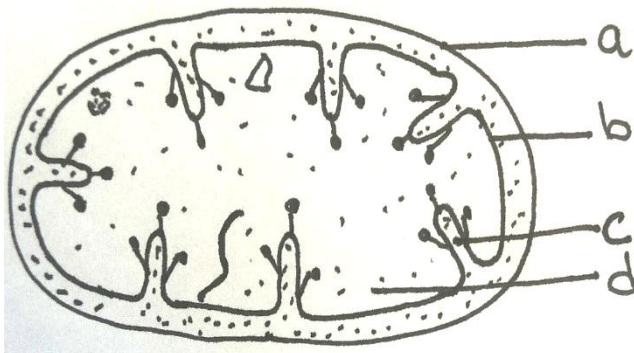
- Which one of the following is involved in the process of autolysis?
a. Lysosome b. Golgi body c. E. R d. Ribosome
- If some solid substances enter a cell through cell membrane, it is called
a. Exocytosis b. Pinocytosis
c. Endocytosis d. Phagocytosis
- A type of negative biotic interaction in which the activity of one organism affects to the other but does not derive any benefit from such activity is
a. Parasitism b. Amensalism
c. Antibiosis d. Allelopathy
- The stem branch which bears flowers is called
a. Pedicel b. Rachis c. Petiole d. Peduncle
- The correct order in hierarchic system of classification is
a. Kingdom-Division-Order-Class-Family-Genus-Species
b. Kingdom- Division-Class- Family-Order-Genus-Species
c. Kingdom- Division-Class-Order-family-Genus-Species
d. Kingdom-Class-Division- Family-Order-Genus-Species
- What type of modified tap root is turnip?
a. Fusiform root b. Napiform root
c. Conical root d. Tuberous root

Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

7. Describe ecological pyramid with reference to pond ecosystem. [4]
8. Study the diagram below and answer the questions. [4]



- i. Name the labelled parts a, b, c and d. [0.5×4 = 2]
- ii. Why this structure is called power house of a cell? [1]
- iii. How 'c' is formed? [1]
9. Describe mutualism and commensalism with suitable examples. [2+2 = 4]
10. What is the name of photosynthetic pigment in plant? Write down the functions of chloroplast. [1+3]

OR

Who proposed five kingdom system of classification. Mention the advantages of five kingdom system over two kingdom system. [1+3]

Group 'C' [BOTANY]

Give long answer to the following questions.

[8 × 2 = 16]

11. Describe the ecological role of light & temperature in flora and fauna in detail. [4+4 = 8]
12. Describe the family Brassicaceae in semi technical terms with floral formula and floral diagram. State the scientific names of any two economic plants of the family. [3+1+3+1 = 8]

OR

Describe life cycle of *Spirogyra* in detail.

[8]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet. [5 × 1 = 5]

13. The main cause of criticism of Darwinism is:
 - a. The atmosphere of earth has changed.
 - b. He has not explained about variations in characters along with heredity.
 - c. Reproduction in animals has been given more importance.
 - d. No explanation about vestigial organs.
14. "During embryonic period, animals repeat the embryonic stages of ancestors". This law is named as
 - a. Mendel law
 - b. Evolution law
 - c. Biogenetic law
 - d. Huxley's law
15. Which is not a locomotory organelle in Protozoa
 - a. Flagellum
 - b. Cilia
 - c. Setae
 - d. Tentacle
16. The period from the time of entry of sporozoites of *Plasmodium* into blood and causing fever is known as
 - a. Instar
 - b. Incubation period
 - c. Grand period
 - d. Inoculation
17. Setae in earthworm is absent in
 - a. First, last and clitellar region
 - b. Second segment
 - c. Ninth segment
 - d. Clitellar region only

Group 'B' [ZOOLOGY]

Give short answers to the following questions. [4 × 4 =16]

18. Draw a neat and fully labeled diagram of life cycle of *Plasmodium vivax* in mosquito. (Description not required).
19. Discuss the transverse binary fission occurring in the *Paramecium caudatum* with well labeled diagram.

20. Given figure is experimental setup of Miller's Experiment.

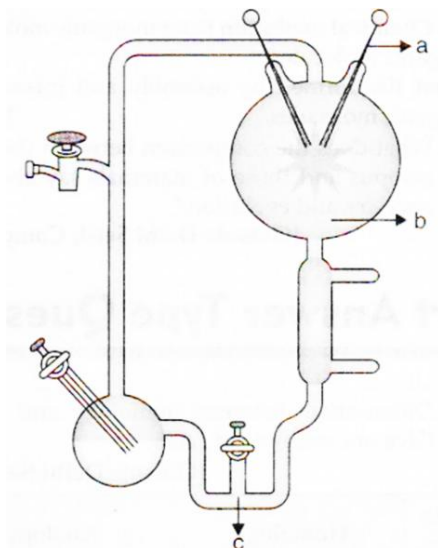
a. Name the label a in the given figure.

b. What gasses are present in the gas chamber.

c. What theory does it support?

d. List the final products of this experiment.

[1+1+1+1]



21. Mention the different external apertures of earthworm.

[4]

OR

Discuss different types of fossils you have studied.

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

22. Define digestion. Describe the structure of the alimentary canal in an earthworm with the help of a well-labeled diagram.

[1+4+3]

OR

What is conjugation? Describe conjugation in *Paramecium caudatum* with labeled diagrams. Mention any two significance of conjugation. [1+4+2+1]

23. Name the theory of evolution purposed by Darwin. Enumerate the key concepts in the evolution theory of Darwin.

[1+7]

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